Oregon Department of Geology and Mineral Industries

Critical Minerals in Oregon

Overview

Securing a resilient supply-chain of domestic **critical minerals and materials** is essential to the economic prosperity and national security of the U.S. In the 21st century, the global need to decarbonize energy production, manufacturing, and transportation is shifting focus from traditional high-value resources (e.g., gold) to a wide range of materials. Oregon has a rich mining history, dating back to gold discoveries in the 1850s. Metal extraction has been dominated by gold, silver, copper, and nickel. Many critical minerals have only become important in the past 50 years, hence they were often discarded with mine waste materials in favor of prioritizing economic mineral targets.

Oregon's historic mine records provide a foundation to guide mineral exploration in the State, now with a critical mineral focus, and to engage with interested parties throughout the process: from initial mapping, to green-energy education and careers, and mine life-cycle management.

Quick Facts

Definition: Critical minerals are non-fuel minerals essential for the U.S. economy and security that are vital for modern technologies, but face a risk of supply chain disruption.

Importance: Critical minerals are fundamental to the transition to a digital and sustainable economy, including:

Batteries

cobalt
lithium
manganese
antimony

Electronics

silicon
gallium
germanium

Defense & Aerospace

beryllium
nickel
chromium

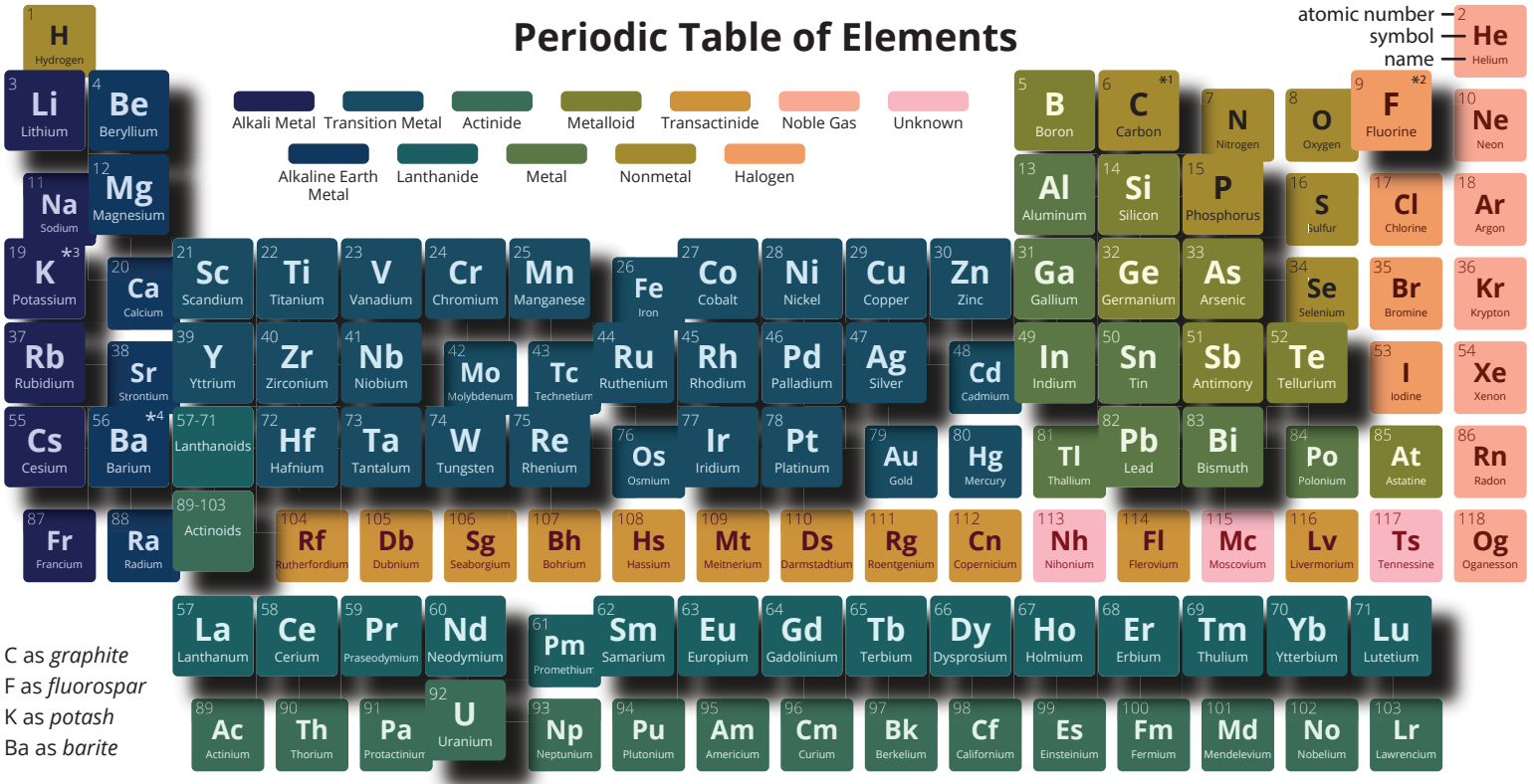
Geopolitics: Resource competition and the potential for export restrictions, highlight the need to secure a domestic supply of critical materials.

Supply: Though found globally, the processing of many critical minerals is concentrated in certain countries, leading to concerns about supply-chain security.

Solutions: Efforts to address vulnerable supply-chains include domestic exploration and mapping, developing new processing technologies, and improving recycling efficiency.

The 2025 Critical Minerals List

The Department of the Interior (DOI) has published lists of Critical Minerals every three years since 2018. As required by The Energy Act of 2020, the U.S. Geological Survey (USGS) has updated the list for 2025. The new list includes 60 minerals, highlighted by the elements in the **Periodic Table** *below*.



The 2025 Critical Minerals Lists sees the addition of 10 minerals, compared to 2022: *boron, copper, lead, metallurgical coal, phosphate, potash, silicon, silver, rhenium,* and *uranium*.

Oregon is home to a unique and varied geology, formed by volcanic and tectonic events that have resulted in a wide range of critical mineral deposits: at least 40 of the 60 critical minerals are indicated within the State in historic records, with others anticipated by common mineral associations.

Contact Information

Jason D. McClaughry, RG
Geological Survey & Services Program Manager
Tribal Liaison

TEL: (541) 519-3419
email: DOGAMI-info@dogami.oregon.gov
web: https://www.oregon.gov/dogami

Critical Minerals In Oregon

Oregon has historically produced, or has known occurrences of, several minerals on the USGS Critical Minerals List. Some of these critical minerals can be found as byproducts of mining other commodities (e.g., gold). According to the latest release of the Mineral Information Layer for Oregon (MILO-4), there are more than 10,000 mine site records for critical minerals identified on the 2025 USGS list (*right*). Key critical minerals in Oregon include:

3 Li Lithium — An essential ingredient in rechargeable batteries. Known occurrences in SE Oregon: McDermitt caldera hosts the largest Li discovery in the U.S.

24 Cr Chromium — Essential in stainless steel and superalloys, for its heat-resistant and anti-corrosion properties. Distributed over large areas of SW and NE Oregon within fragments of former ocean crust.

27 Co Cobalt — Essential in rechargeable batteries and alloys. Found in localized deposits in NE Oregon and as distributed deposits with Ni and Cr in laterite soils.

28 Ni Nickel — In high demand for clean energy technologies. Found in NW and SW Oregon as distributed deposits with Co and Cr, largely in laterite soils formed from uplifted and accreted oceanic rocks.

47 Ag Silver — A key element in the energy transition for solar panels, EVs, and advanced electronics. Ag has been mined extensively in NE and SW Oregon.

29 Cu Copper — Used in electronics and needed in increasing quantities for the energy transition. Cu has been mined extensively Oregon, particularly in NE and SW as a coproduct with gold and silver.

74 W Tungsten — Essential for cemented carbides, cutting tools, and specialized alloys. W is found in NE Oregon, in association with Au, Mo, and Cu.

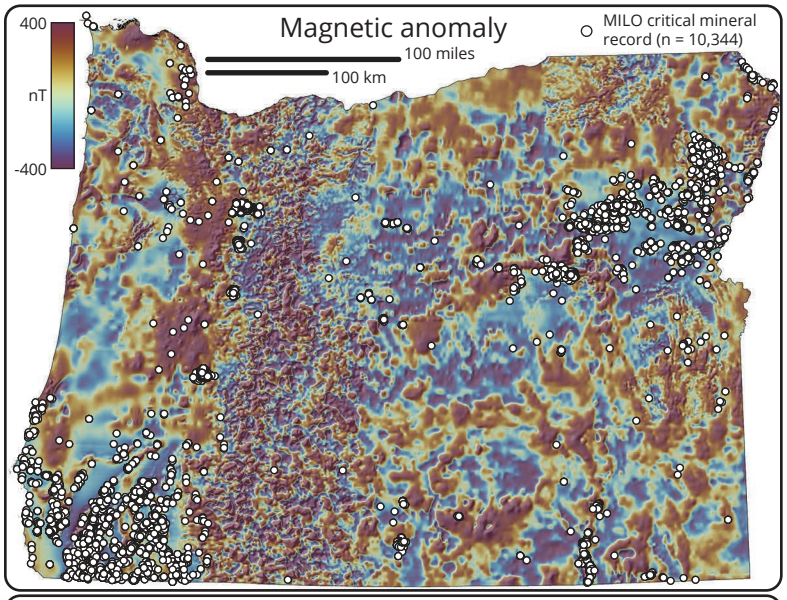
51 Sb Antimony — Essential to national security, defense, and modern technologies. Occurrences in central and NE Oregon, formed in epithermal systems: deposited by hot, mineral-rich volcanic fluids.

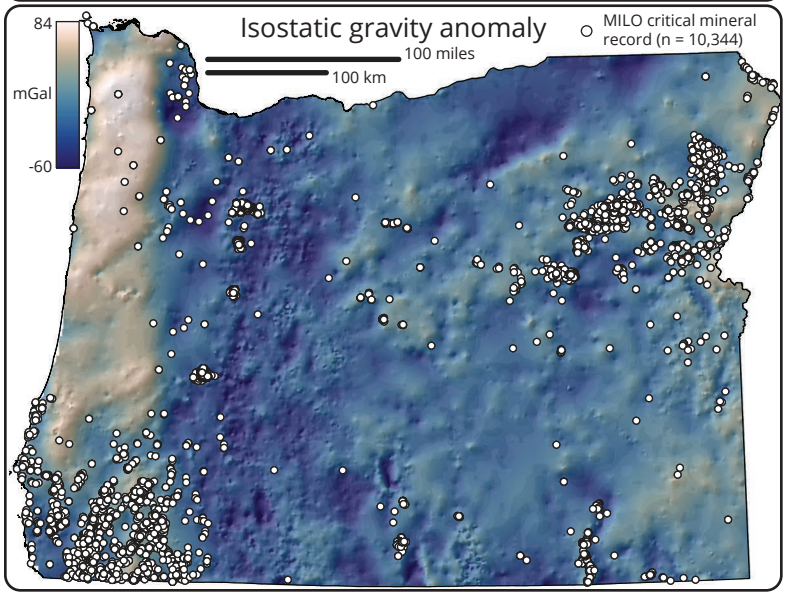
13 Al Aluminum — Used extensively in key sectors like construction, transportation, and national defense. The ore *bauxite* is found in NW Oregon.

Next Steps

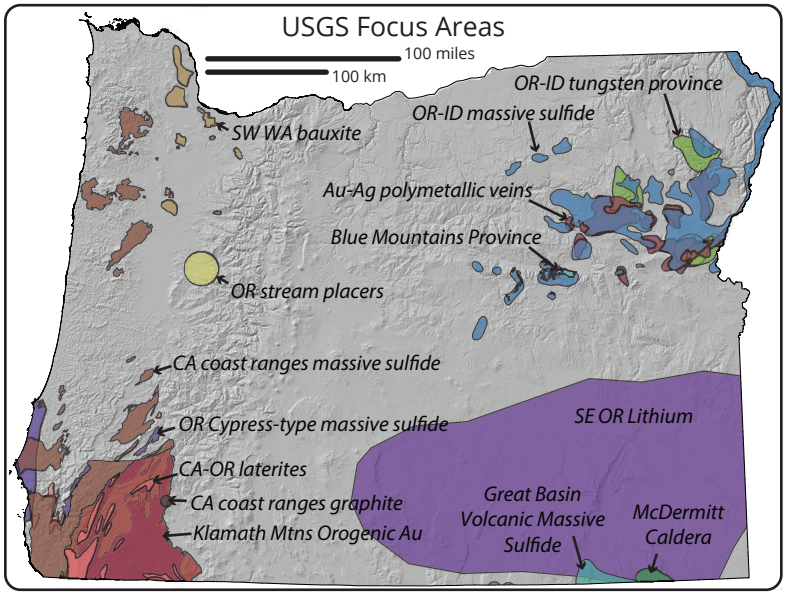
DOGAMI is engaged in critical minerals exploration across the State, with active resource characterization projects in McDermitt Caldera, SE Oregon (Li), and the Blue Mountains (Bi, Co, Ga, Ge, W), and USGS geophysical data collection scheduled for 2025-2026 in NE Oregon. DOGAMI will use the MILO-4 dataset, together with arising data, to define new USGS focus areas and plan future exploration targets.

DOGAMI's research and exploration of critical minerals is ***guided by outreach and engagement*** with communities and Tribal Nations at every step of the process. By putting community collaboration at the forefront of the research and development process, DOGAMI will fulfill its ***mission of providing earth science information and regulation to make Oregon safe and prosperous***.





Critical mineral occurrences (white dots) in Oregon are closely related to gravity and magnetic anomalies, representing geologic features such as rock contacts and faults, below Earth's surface.



USGS focus areas are defined by known occurrences of critical minerals: DOGAMI is working to update areas using emerging data, and geologic characterization from mapping actiities.

Oregon Department of Geology and Mineral Industries

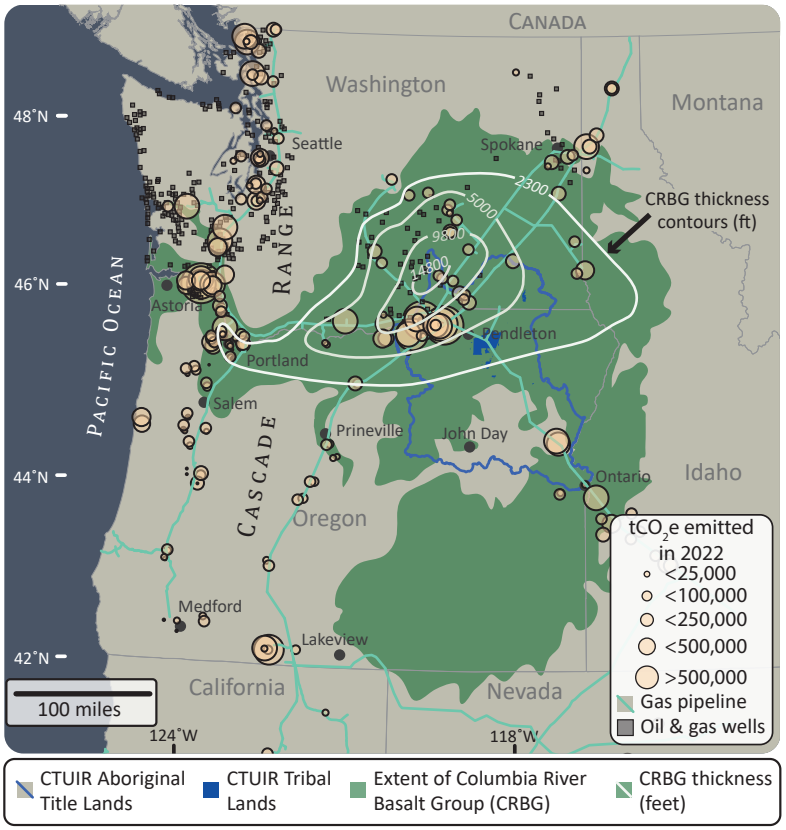
Geologic Carbon Sequestration in Oregon
An Update on State Investigations: September 2025

Project Overview

Oregon has the potential to develop a thriving carbon-negative economy, attracting innovative technologies like direct air capture, point-source removal, and geologic carbon sequestration, that capitalize on Oregon's Climate Protection Program, robust renewable energy grid, and widely distributed geologic storage system.

The State of Oregon has established ambitious climate goals aimed at significantly reducing greenhouse gas emissions by 2035, and achieving net-zero carbon emissions by 2050. Realizing those goals will require a multi-faceted approach. Establishing regional carbon-management hubs across the State, capable of safely and permanently storing carbon dioxide (CO_2) could play an important role.

The State has directed the Department of Geology & Mineral Industries (DOGAMI) to investigate the potential for geologic carbon sequestration in the Columbia River Basalt Group (CRBG, as shown on the right).



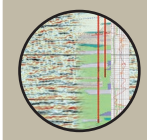
The Columbia River Basalt Group (CRBG)

Near-term Project Goals



Community Engagement

- Listening to Tribal Nations, regional stakeholders, and nearby property owners
- Educational program development
- Assessment of community workforce and economic needs



Technical Feasibility

- Install 1-3 stratigraphic wells on Department of State Lands (DSL) property
- Surface-based geologic mapping
- Subsurface geophysical mapping
- Hydrologic and chemical data collection
- Seismic-hazard analysis



Regulatory Structure

- Investigation of Class VI underground injection well (UIC) regulatory primacy
- Development of permitting pathways for carbon sequestration and geothermal exploration.

Key EPA Regulations for Class VI Carbon Dioxide Injection Wells

Protecting groundwater and other natural resources is of the utmost importance to any carbon storage project. The US Environmental Protection Agency (EPA) has strict standards under the *Clean Water Act*, that must be met for any carbon injection to be permitted, including:

1. Geologic characterization of the storage reservoir and identification of a robust seal
2. Water quality and aquifer protection assessment
3. Well construction requirements, and well materials corrosion monitoring
4. Long-term financial liability and stewardship requirements
5. Community engagement, including a period of public consultation prior to permitting

Next Steps for the Project

1. Establish agreements with academic, workforce, and community partners
2. Engagement and outreach with all potentially affected communities
3. Stratigraphic well site-selection and well installation
4. Geologic models using surface and underground data
5. Development of state regulations
6. Legislative considerations

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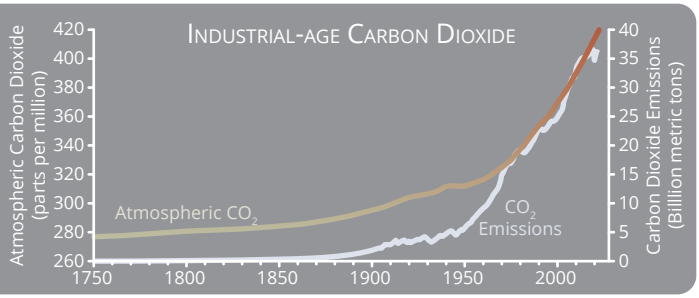
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Geologic Carbon Storage in the Columbia River Basalts: Some Key Points

Climate Change: The Need For Action

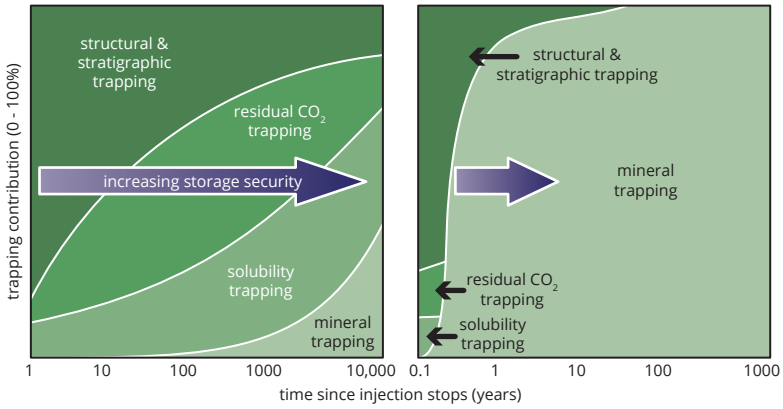
1.3°C	rise since pre-industrial era
2.0°C	target set in the Paris Climate Agreement
800 $GtCO_2$	remaining budget to exceed targets
40 $GtCO_2$ / year	in global emissions
20 years	remaining to meet targets



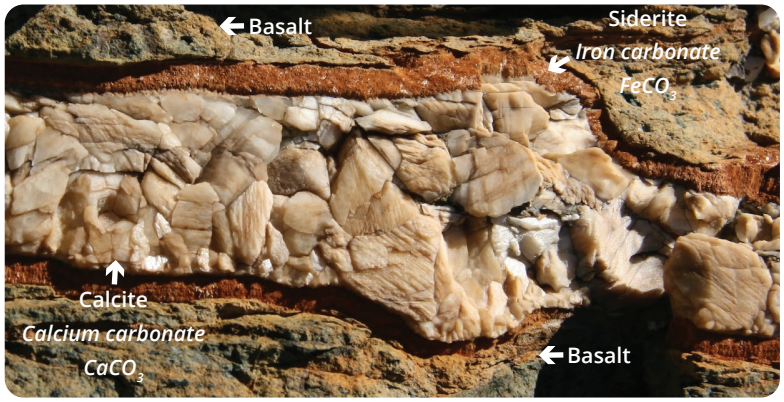
What is mineral storage?

Geologic carbon storage (GCS) is the process of capturing carbon dioxide (CO_2) from industrial sources or the atmosphere, compressing it into a fluid, and injecting it deep underground into suitable geological formations to be permanently stored.

Most types of GCS system store CO_2 as a fluid in the subsurface, and require long-term monitoring to ensure that fluid does not leak in the future. However, CO_2-rich fluids will react with the minerals in basalt, turning that CO_2 into stone over time (*mineral trapping*). This process occurs slowly in nature, and is accelerated with the addition of higher CO_2 concentrations in the fluid.



Types of Geologic Carbon Storage (GCS)



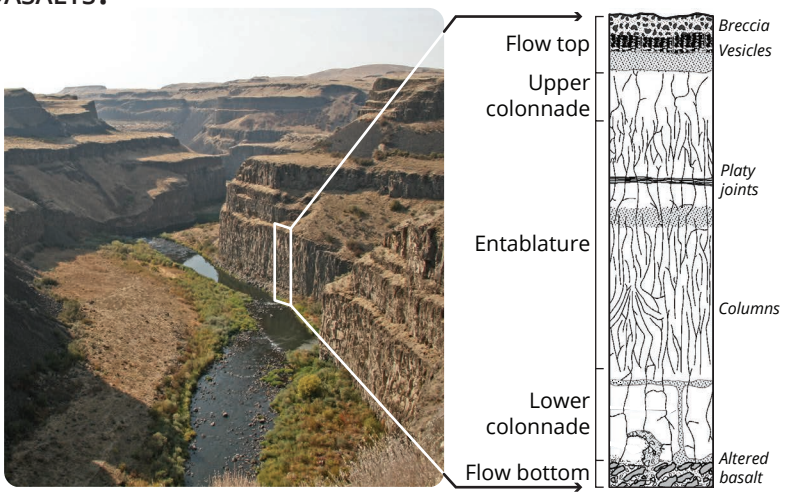
Natural carbon mineralization in basalt

Where could CO_2 be stored in the Columbia River Basalts?

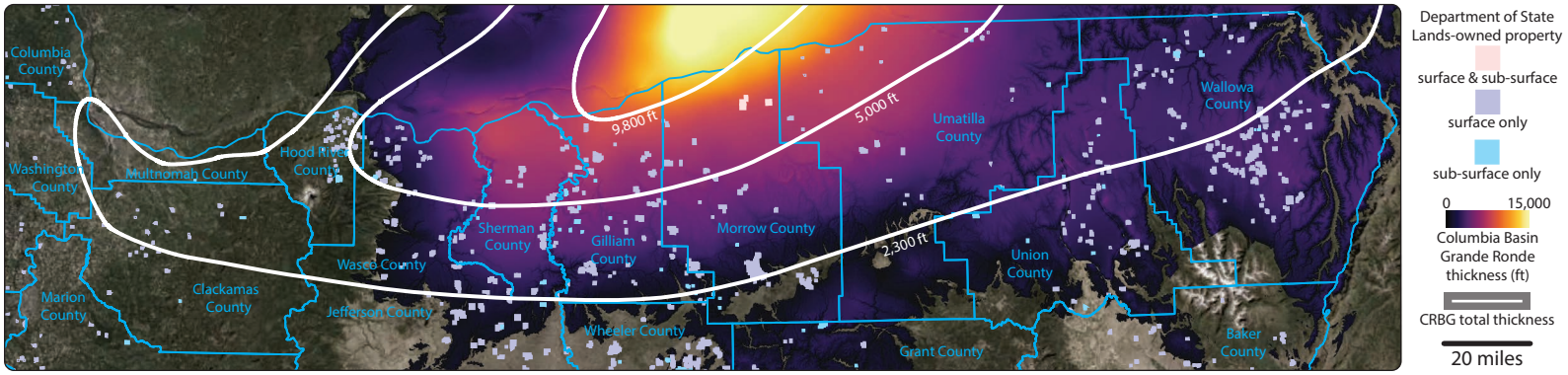
Potential CO_2 reservoirs in the CRBG reside in the *Grande Ronde Basalt*, far below the potable water zone in the Columbia Basin. The Grande Ronde includes ~100 individual lavas, with a total thickness up to 14,760 feet and covering ~65,600 square miles. It is estminated that the Grande Ronde may be capable of storing 40 billion metric tons (Gt) of CO_2.

Grande Ronde lavas consist of fractured, weathered, fragmented, and vesicular top and base zones, and denser interior zones that often have colonnade joints. Lava tops and bottoms can have porosity up to ~40%, whereas unit interior porosity is typically 1-3%. Lava tops and bases may be good reservoirs for CO_2 storage, with lava interiors acting as seals.

The *stratigraphic research well* will allow characterization of rock properties, to determine the suitability of the basalts for CO_2 mineral storage, at great depth.



Lavas of the Columbia RIver Basalt Group



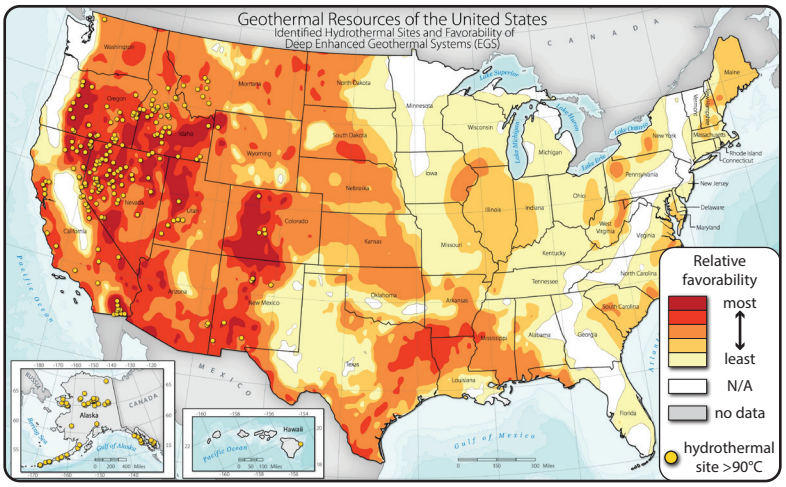
Thickness of the Grande Ronde Basalt within the Columbia River Basin. A Department of State Lands property in northern Oregon will be selected for the stratigraphic research well. ***No CO_2 will be injected during this project.***

Oregon Department of Geology and Mineral Industries

Geothermal Energy in Oregon

Overview

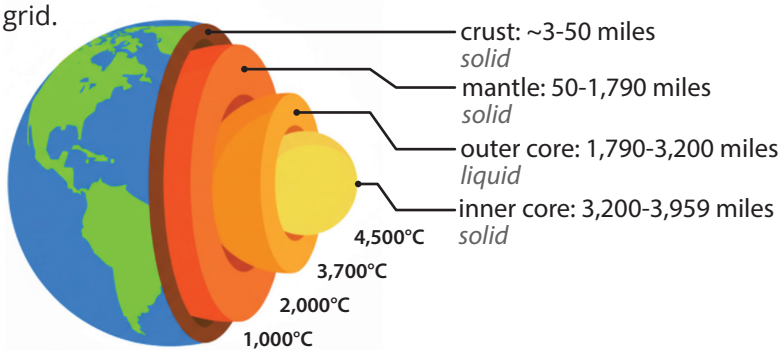
Geothermal energy is a crucial potential source of constant, 24/7 renewable power in the Western U.S. This potential is due to the region's active geology, marked by tectonic plate boundaries, volcanic activity (such as the Cascade Range), and hot spot areas. While California and Nevada currently lead the nation in installed geothermal electricity capacity, Oregon is one of the few states with existing utility-scale geothermal power generation and consistently ranks among the top states for geothermal potential. With known resources in the Cascade Range and the Basin and Range Province of its central and southeastern counties, Oregon is well positioned to leverage both conventional (hydrothermal) and next-generation technologies, such as Enhanced Geothermal Systems (EGS), to expand this reliable resource and help the entire Western grid transition to a clean-energy future.



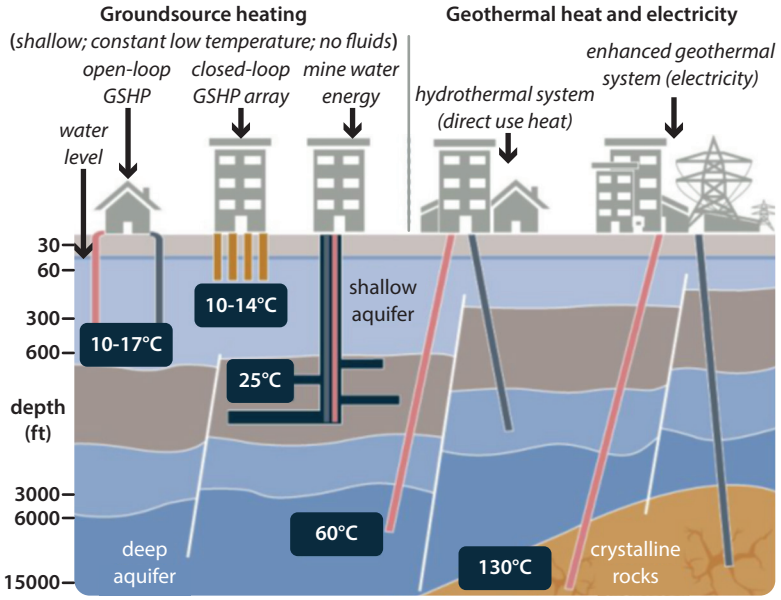
National Renewable Energy Laboratory (NREL) map of geothermal resources in the U.S. highlighting Oregon's geothermal potential.

What is Geothermal Energy

Definition: Geothermal is a renewable energy resource that taps into the heat beneath the Earth's surface. Drilled wells can extract hot fluids (water or steam), which are used to generate *electricity* or provide *direct heating*. Unlike solar or wind, geothermal is not dependent on weather or time of day, making it a reliable, constant (*baseload*) power source for the grid.



Types: Geothermal energy is used in two main ways: *Electricity generation* uses high-temperature fluids to spin turbines; *Direct Use and district heating* use lower temperature hot fluids to heat homes, businesses, and pools, displacing fossil fuels and saving money. Oregon is a leader in both applications.



New technologies: *Enhanced Geothermal Systems* (EGS) and *Superhot Rock Geothermal* could help access areas of hot, dry rock where natural fluids are lacking, significantly expanding Oregon's resource potential (e.g., Newberry Caldera projects).

Why Geothermal Matters

Reliable Baseload Power: Unlike solar and wind, geothermal runs 24/7, providing system stability.

Low Emissions & Clean Energy Goals: Helping Oregon to meet its goals of 100% clean electricity generation by 2040.

Small Land Footprint: Requires significantly less land per megawatt than other renewables.

Economic Impact: Provides local jobs in rural communities and stable energy costs.

Barriers and Challenges

High Upfront Costs: Exploration and drilling can be time-consuming and expensive.

Resource Risk/Uncertainty: Currently there are large data gaps for assessing geothermal resources (heat and flow).

Siting and Permitting: The need for careful siting to avoid impacts on environmental areas (e.g., hot springs).

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Geological Survey & Services Program Manager
Tribal Liaison
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email: DOGAMI-info@dogami.oregon.gov
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Geothermal Energy In Oregon

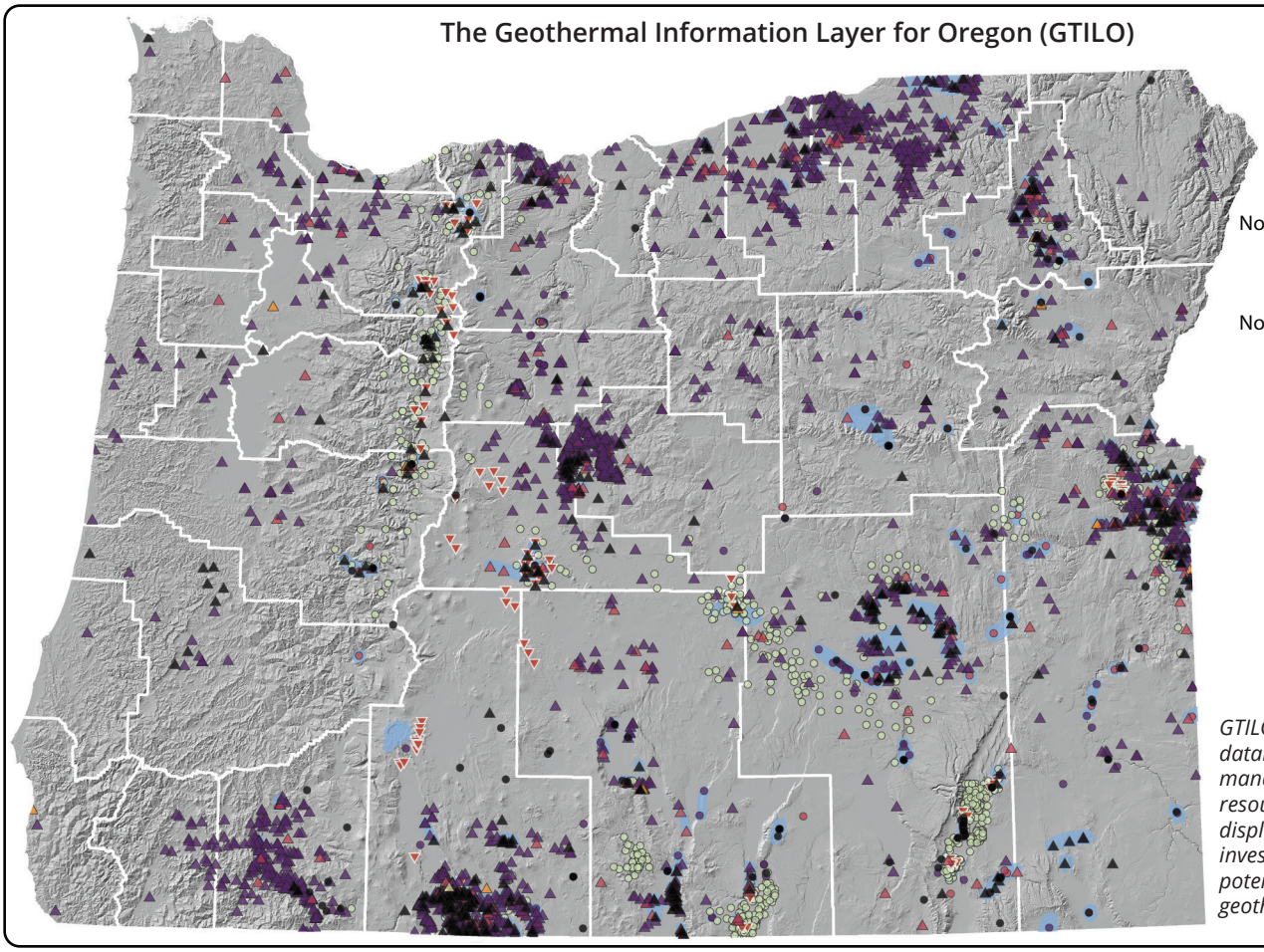
Oregon holds significant geothermal assets associated with volcanic features such as the Cascade Range and High Lava Plains, as well as systems associated with high heat flow caused by crustal extension and thinning in the Basin and Range Province of southeastern Oregon.

Oregon hosts utility-scale geothermal power plants, such as the Neal Hot Springs facility in Eastern Oregon. Oregon is a national example of *Direct Use*, notably in Klamath Falls, which for decades has used geothermal heating for its downtown buildings and the Oregon Institute of Technology. Recently, the Newberry Geothermal System has become globally significant as a premier site for *Enhanced Geothermal Systems* research, demonstrating *Superhot Rock* technology can economically and reliably tap extreme heat from deep, dry rock.



Mazama Energy operations in Newberry, OR, where a temperature of 331°C has been reached.

The Geothermal Information Layer for Oregon (GTILO)



Low Temperature Geothermal Resource Areas

Hot & Warm Springs (690)
No data · Low · High

Low Temperature Wells (4,344)
No data · Low · High

Geothermal Prospect Wells (1,019)

Geothermal Well (66)

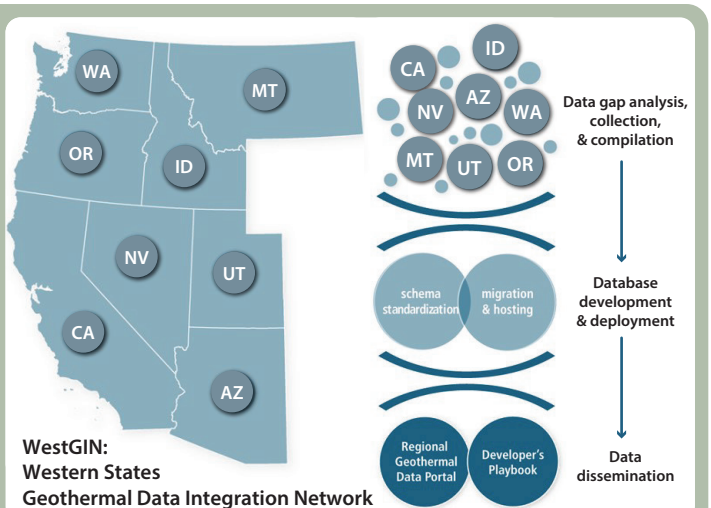
50 Miles
50 Kilometers

GTILO is DOGAMI's geospatial database system that stores and manages Oregon's geothermal resource information. This map displays data that either have been investigated for geothermal resource potential or show potential for a geothermal resource.

Next Steps

The western U.S. holds substantial untapped geothermal energy potential, but development is often hindered by inaccessible data. Public and private geothermal datasets are scattered across state geological surveys, private companies, and universities in various formats. This impedes regional analysis, elevates exploration risks, and discourages investment, limiting geothermal expansion.

DOGAMI is joining with state surveys in NV, WA, UT, ID, MT, AZ, and CA to develop a unified, publicly accessible, interactive geothermal data platform known as **WestGIN**. The project will streamline data access, build collaborations among state surveys, prioritize data collection and dissemination, and create regional products from data synthesis including geothermal playbook chapters. In doing so, DOGAMI will fulfill its ***mission of providing earth science information and regulation to make Oregon safe and prosperous***.



WestGIN:
Western States Geothermal Data Integration Network

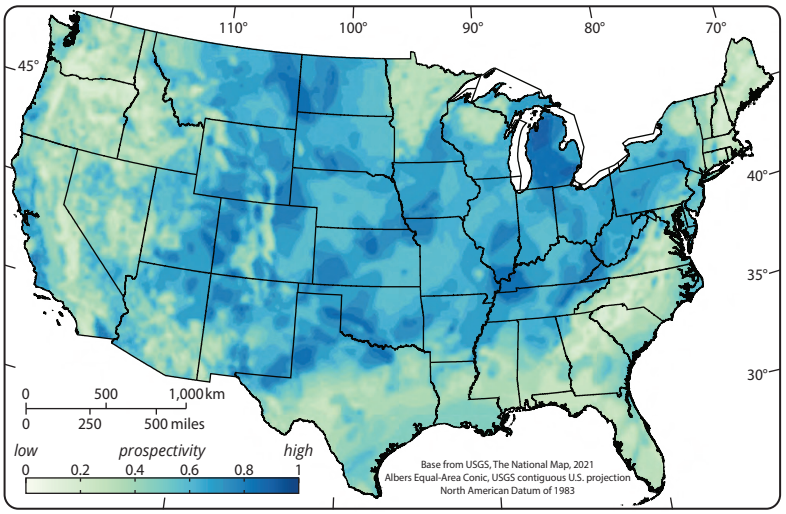
Oregon Department of Geology and Mineral Industries

Geologic Hydrogen in Oregon

atomic number – 1
H – symbol
name – Hydrogen
1.007 – atomic mass

Overview

Secure and reliable energy is critical to U.S. national security, especially as the demand for clean fuels used in electric power generation, manufacturing, and transportation become a priority. Factors including population growth, the increasing energy needs of data centers, reshoring of manufacturing, and the electrification of transport and other sectors are straining our ability to meet energy demands. To address this, the U.S. is exploring the potential of clean, low-carbon hydrogen as a new, abundant, and domestic energy source. Hydrogen is a versatile fuel suitable for transportation, industrial processes, and power generation because it produces zero greenhouse gas emissions at the point of use. The goal is to make hydrogen a cost-effective and readily available option to reduce carbon emissions and achieve a cleaner energy future.



Median (P50) hydrogen prospectivity[1] in the contiguous U.S.

What is geologic hydrogen

Geologic hydrogen (sometimes referred to as *white* or *gold* hydrogen) is sourced or released from underground rocks by natural processes, and is generated naturally by several processes (*right*).

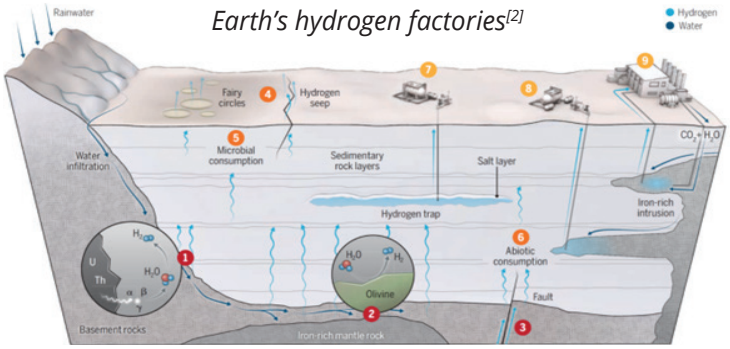
- **Serpentinization:** water oxidizes iron-rich minerals in rocks (e.g., olivine and pyroxene). Oxygen from the water binds with iron in the rock, leaving behind free hydrogen.
- **Radiolysis:** natural radiation from radioactive elements present in rocks, breaks down water molecules (H2O), to produce free hydrogen.
- **Biological activity:** hydrogen produced by microbial life during metabolism (metabolic hydrogen or biohydrogen), such as through fermentation or nitrogen fixation.
- **Thermal cracking of hydrocarbons:** organic matter deep within crustal basins break down to form hydrogen.

These natural processes can be replicated and engineered to increase hydrogen production rates, such as production from fossil-fuels using a process known as steam methane reforming. This process uses large amounts of energy and produces substantial amounts of carbon dioxide as a by-product. Hydrogen can also be produced by electrolysis of water, which can have a very low environmental impact if using 100% renewable or nuclear energy sources.

"color"	Black/Brown Hydrogen	Gray Hydrogen	Blue Hydrogen	Yellow Hydrogen	Purple Hydrogen	Green Hydrogen
feedstock	Coal	Fossil fuels	Fossil fuels	Water	Water	Water
production process	Gasification	Steam (methane) reforming	Steam (methane) reforming with CCUS	Electrolysis using solar power	Electrolysis using nuclear power	Electrolysis using renewable electricity
products	H_2 & CO_2	H_2 & CO_2	H_2 & CO_2 (captured)	H_2 & O_2	H_2 & O_2	H_2 & O_2

The engineered-hydrogen "rainbow" [3]

Earth's hydrogen factories[2]



Generation
1. Radiolysis
2. Serpentinization
3. Deep-seated

Loss mechanisms
4. Seeps
5. Microbes
6. Abiotic reactions

Extraction
7. Traps
8. Direct
9. Enhanced

Current & emerging uses of hydrogen

Transportation: used in hydrogen fuel cells to power electric vehicles with zero tailpipe emissions

Industrial: a key ingredient in ammonia, for fertilizers, cleaning products, and refrigerants

Energy Storage: improving grid stability for energy produced by intermittent sources (e.g., solar, wind)

Aerospace: used as a rocket propellant and potentially in aviation fuel for longer-range aircraft

Industrial: replacing coal as both the source of heat and the reducing agent in steel making

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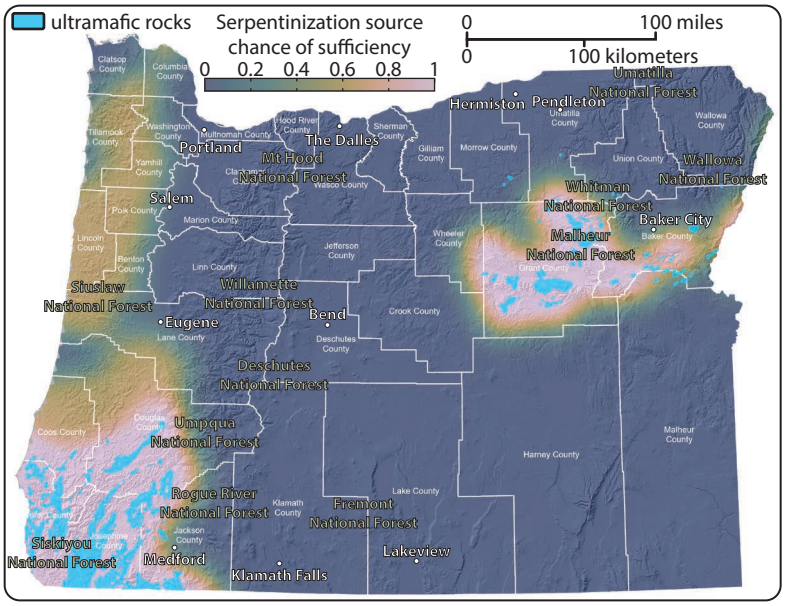
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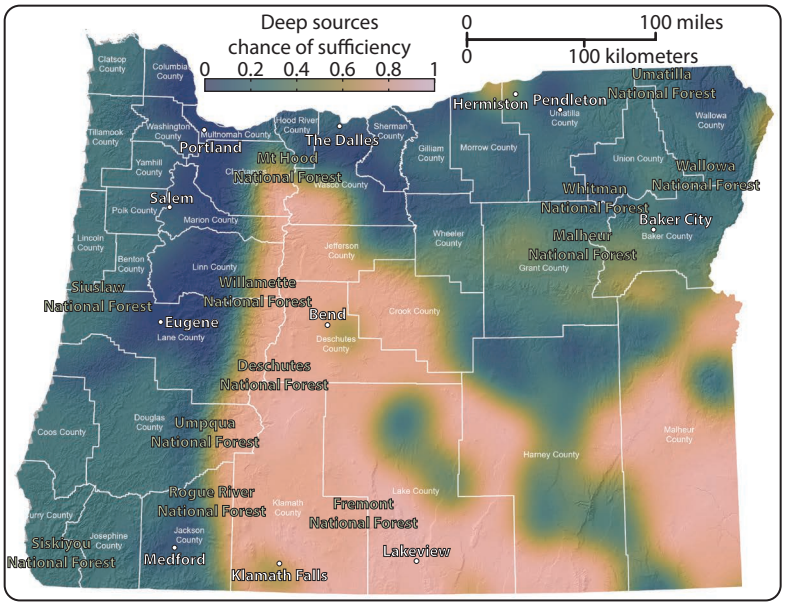
Hydrogen In Oregon

Recent U.S. Geological Survey (USGS) reports[1] suggest geologic hydrogen exists in quantities large enough to make a significant contribution to the U.S. energy portfolio (*see overleaf*). USGS prospectivity models suggest that Oregon has a high potential for hydrogen **sources**, both at the surface and at depth (*below*). Those models provide a useful basis for early exploration work in Oregon and can be used as a guide for future geologic mapping and rock characterization specifically for geologic hydrogen potential.

One of the primary natural source mechanisms for hydrogen production is ***serpentinization***: the hydration of iron–bearing minerals, such as olivine and pyroxene, in *mantle peridotites* (e.g., *right*). Serpentinization is a "redox" (oxidation-reduction) reaction, in which the ferrous iron (Fe^{2+}) is oxidized to ferric iron (Fe^{3+}), and water (H_2O) is reduced to release molecular hydrogen (H_2). This released hydrogen can be stored in the subsurface if there is a seal and reservoir present, it may react with subsurface compounds and minerals, or it can migrate upward toward Earth's surface. The USGS maps therefore indicate probability of the *ingredients* for hydrogen, but are not a guarantee that hydrogen will be present.



Serpentinization-source *hydrogen potential in Oregon*



Deep-source *geologic hydrogen potential in Oregon*



Serpentinized ultramafic rocks within a shear zone, Klamath Mountains, Josephine County, Oregon.



Serpentinized ultramafic rock with high-iron content, Strawberry Mountains, Grant County, Oregon.

Next Steps

In DOGAMI's role as a ***regulatory and geological survey***, we intend to facilitate the research and development of geologic hydrogen resources in Oregon. The next steps would be to collect geophysical data, such as gravity, magnetic, and electromagnetic data to characterize the shallow subsurface and evaluate the potential for ***hydrogen sources, reservoirs, and seals*** across the state. Data collected by drilling information holes could be used to further evaluate subsurface conditions. Notably, exploration is classified as an outright land use in Oregon, and exploration projects need only landowner approval to proceed to permitting.

DOGAMI's research and exploration of geologic hydrogen resources will be ***guided by outreach and engagement*** with communities and Tribal Nations at every step of the process. DOGAMI will work with potentially impacted parties to understand concerns and perspectives, and inform ongoing project planning while openly and directly addressing community feedback.

By putting community collaboration at the forefront of the research and development process, and incorporating industry-established best practices, DOGAMI will fulfill its ***mission of providing earth science information and regulation to make Oregon safe and prosperous***.

References

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